Oscillator-Centric Analysis of Power Grid Stability

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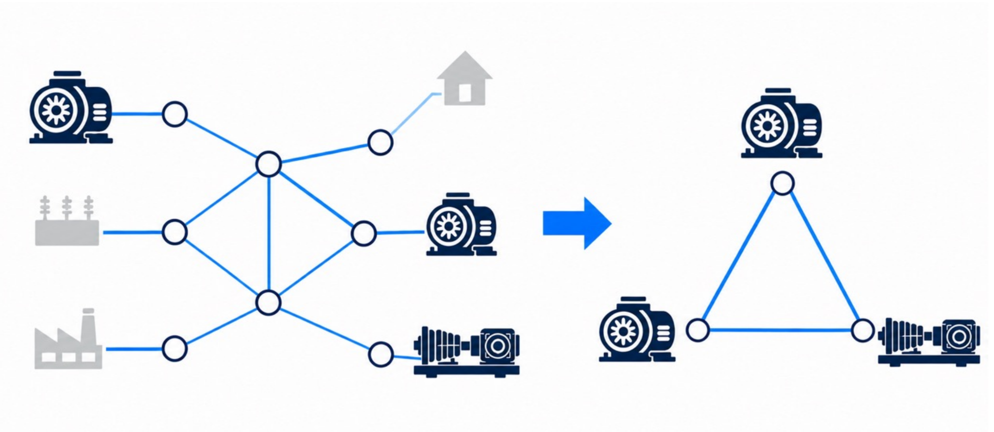
01 Introduction



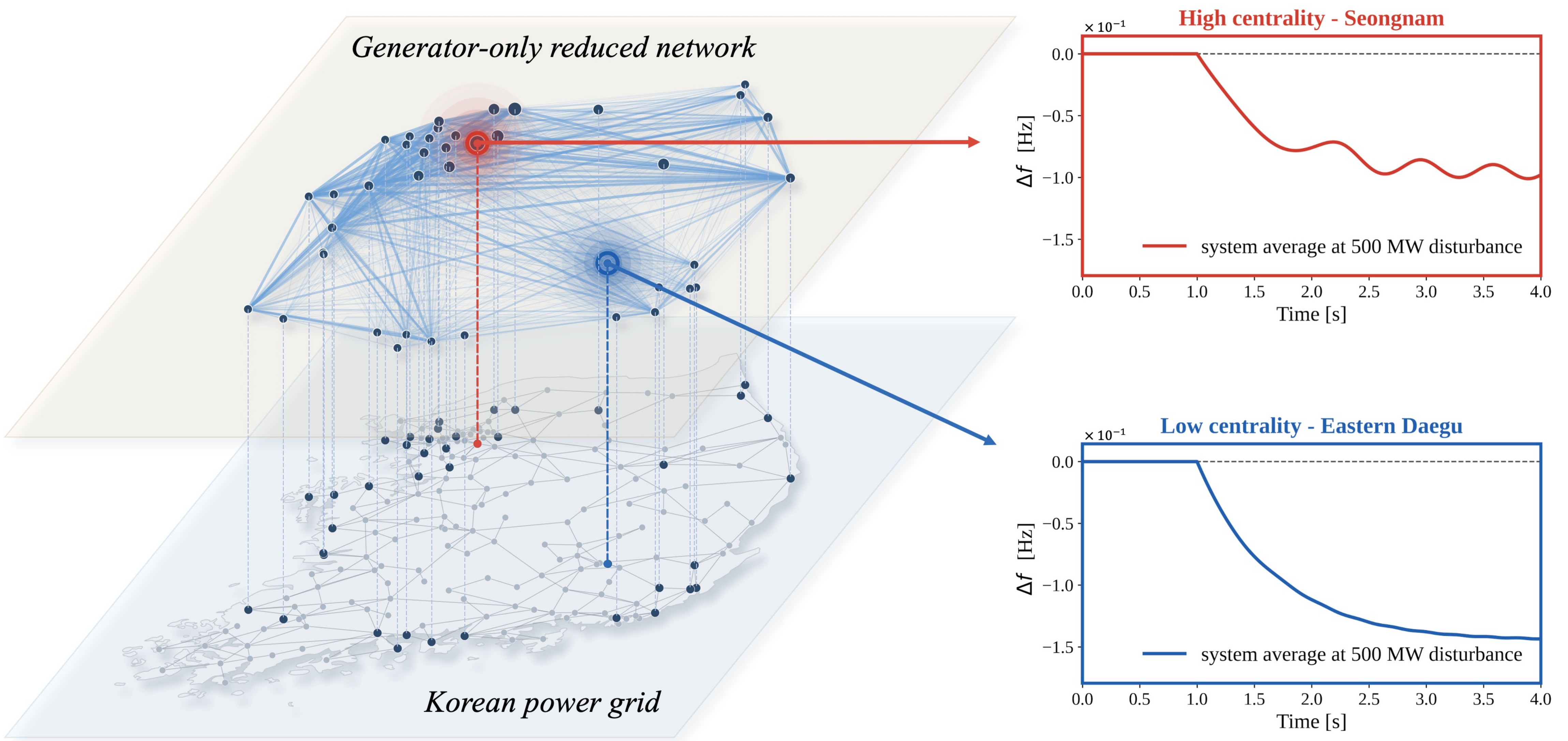
Power grids are networks of interacting components. Some network studies simplify the network structure or node properties to reveal general synchronization principles. However, these simplifications may obscure the physical features of real power grids.

Our question:

How can we analyze power-grid dynamics as an oscillator network while preserving its physical features?

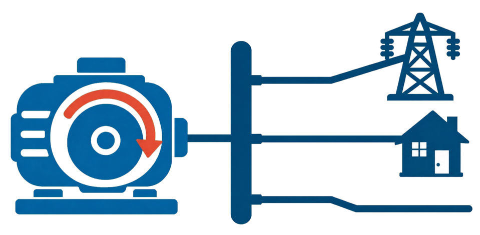
We model only generators as dynamic nodes and load buses are eliminated algebraically, while their electrical effects remain as effective coupling between generators. The resulting reduced oscillator network preserves the dynamics and heterogeneity of the power grid.

02 Modeling framework



Network reduction preserves effective generator-to-generator coupling while removing inertia-free buses. Even under the same 500 MW disturbance, a highly central generator and a peripheral generator exhibit different frequency responses, highlighting the role of network position.

03 What are the physical oscillators in the grid?



A bus is an electrical junction, and inertia belongs to connected devices. We model only generators dynamically, eliminating buses while retaining their effect on generator coupling.

04 Network modeling and dynamics

- Rotor dynamics and power transmission

- Power imbalance drives rotor dynamics.

$$M_i\ddot{\delta}_i + D_i\dot{\delta}_i = P_i^m - P_i^e$$

- Electrical power transfer depends on phase-angle differences and coupling strengths.

$$M_i\ddot{\delta}_i + D_i\dot{\delta}_i = P_i^m + \sum_j K_{ij} sin(\delta_j - \delta_i)$$

- Coupling after Kron reduction

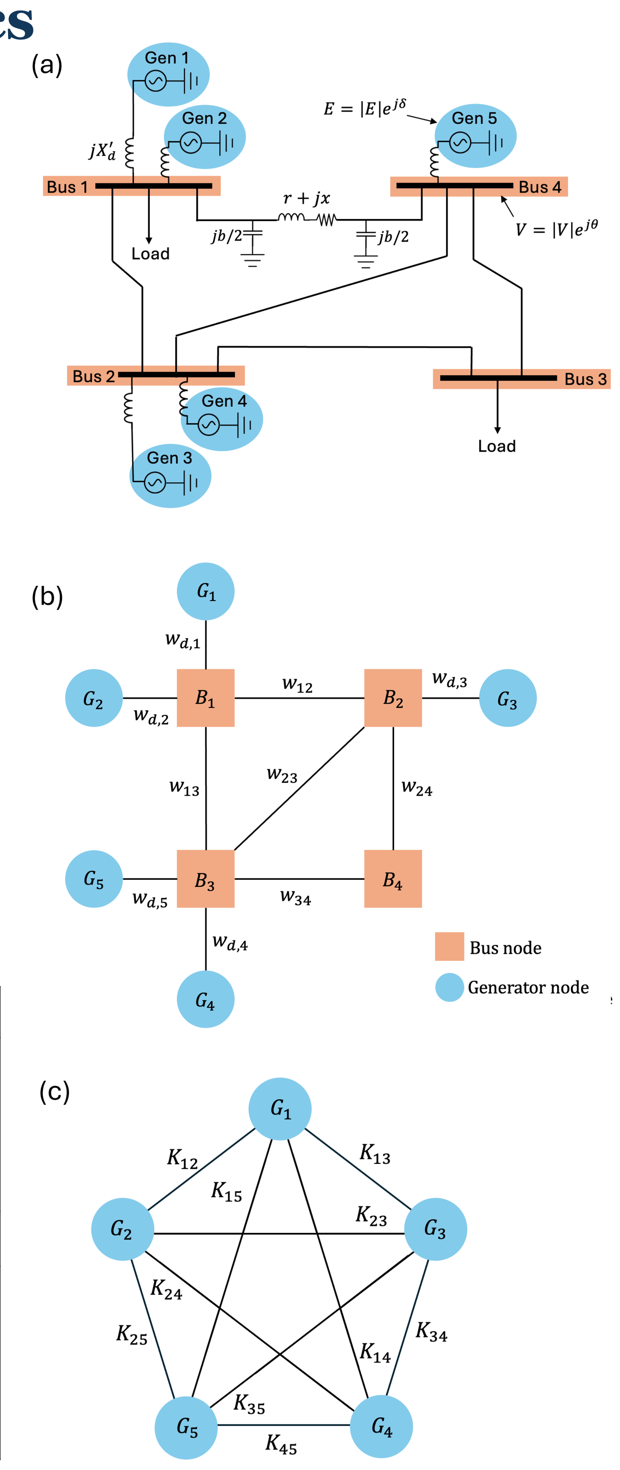
- Reduction removes buses but preserves electrical paths.
- The reduced matrix defines heterogeneous K_{ij}

- Generator-only oscillator network

193 inertia-free buses are removed from the state vector
103 generator units interact through network

What we preserve, perturb, and measure

Keep the physics	Disturb the grid
Generator inertia M_i Topology-dependent coupling K_{ij} KPG-193 operating point	Power loss at each of 103 generators 100 · 200 · 500 MW
Measure the response	
Rate of change of frequency (RoCoF) : $\frac{d\Delta f_i(t)}{dt}\Big\rvert_{t=t0}$ Frequency Nadir : min t ∈ $[t_0, T]$, $\Delta f_i(t)$	



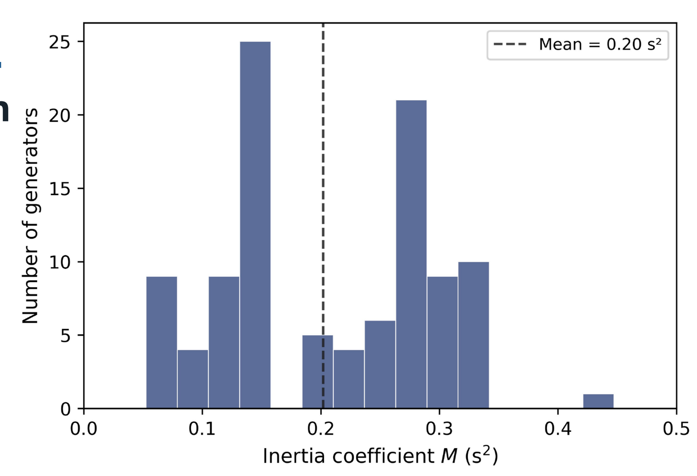
05 Node heterogeneity shapes the initial response

Generators have heterogeneous physical properties.

Each oscillator's inertia M determined by its **generation type and capacity.** $M_i = 2H_iS_i / \omega_s S_{base}$

Generator inertia governs initial frequency response

The inertia of the disturbed generator shows an almost **perfect negative rank correlation** with the system average RoCoF (Spearman $\rho = -0.99$).



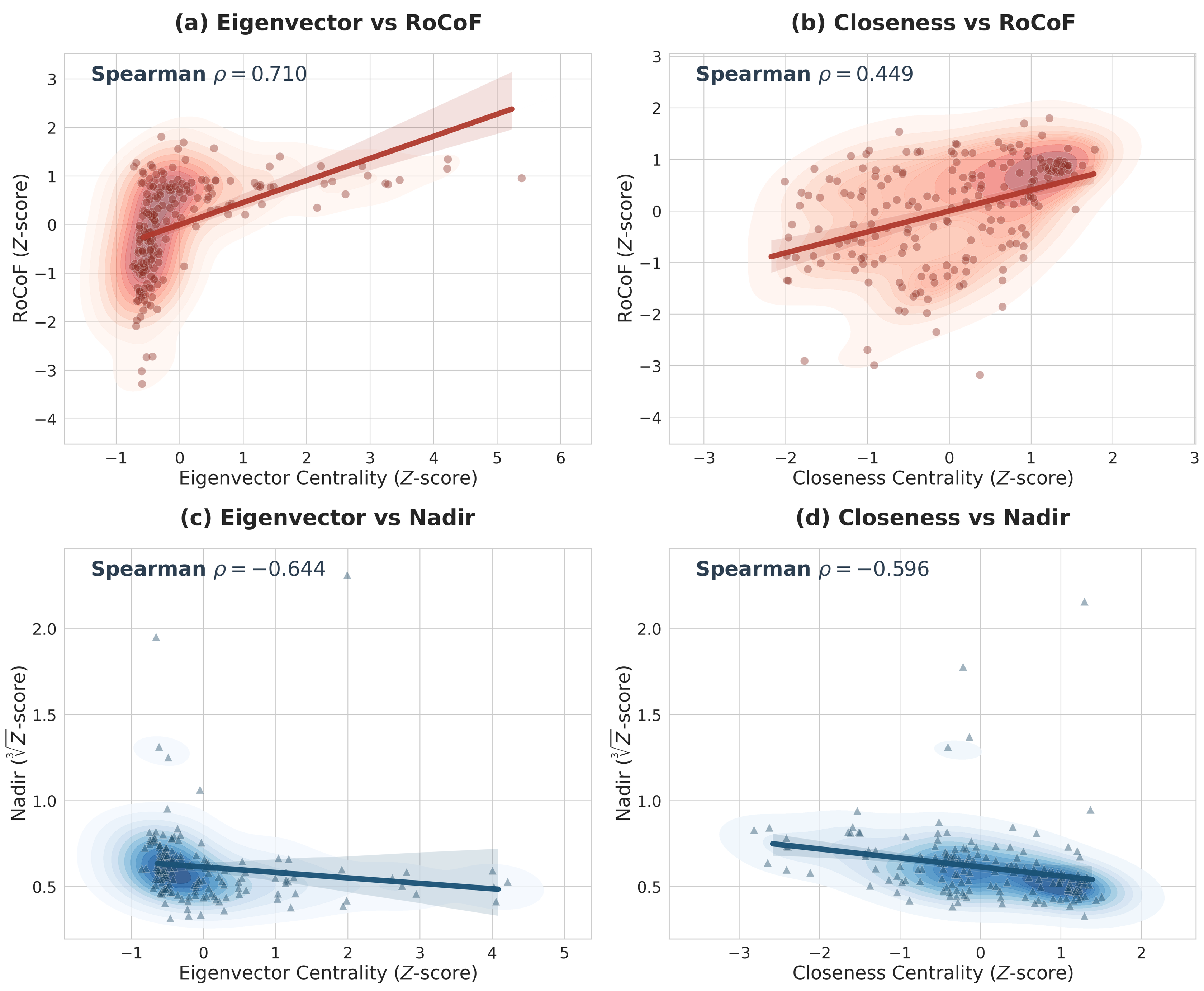
However, the correlation with the nadir was weaker than RoCoF.

500 MW disturbance applied separately to five generators with **identical inertia**, the resulting frequency nadir differed by up to **25%.**

This variation suggests that factors other than inertia also influence the response.

Gen ID	Bus ID	Nadir (Hz)	RoCoF (Hz/s)
102	147	0.0963	0.1056
41	49	0.0868	0.1045
56	79	0.0847	0.1041
26	33	0.0774	0.1027
21	26	0.0769	0.1026

06 Coupling heterogeneity reveals two network roles



Local vulnerability · Receiver view (top)

Each red point represents one bus. Its median RoCoF over all disturbance locations and sizes is compared with its centrality.

A **positive correlation** means that central buses tend to experience deeper fluctuation.

Global impact · Source view (bottom)

Each blue triangle represents one generator disturbance scenario. The frequency nadir averaged over all buses is compared with the disturbed generator's node centrality.

A **negative correlation** means that a central source spreads the imbalance more broadly, reducing the grid-wide frequency drop.

07 Conclusion and discussions

- Our oscillator-centric model retains only generators as dynamic nodes. Load bus states are eliminated algebraically while their electrical effects remain.
- Generator inertia governs the initial response, while network centrality identifies sensitive receivers and broadly spreading disturbance sources.

Limitations and future work

- The model focuses on inertia and therefore simplifies damping. Treating those two independently could reveal richer dynamics.
- Including machine controls and time-varying demand would better reflect real grid operation.
- Future work will validate these extensions against detailed simulations or event data.

M. Kim and H. Kim, "Frequency synchronization analysis of an oscillator-based power grid," New Phys.: Sae Mulli 76(8), in press (2026).

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